

Ali Ahmed

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EDUCATION

Alexandria University, Faculty of Computing and Data Sciences
Bachelor's in Data Science

Alexandria, EG
June 2027

- Relevant Coursework: Machine Learning, Probability & Statistics, Discrete Mathematics, Data Visualization, Software Engineering, Design & Analysis of Experiments, Survey Methodology, and Introduction to AI, stochastic.

PROFESSIONAL EXPERIENCE

DEPI (Ministry of communication certification)

Alexandria, EG
Oct 2024 – May 2025

Data Science Trainee /Team leader

- Completed web-scraping, machine-learning, and database projects, applying data-science workflows to practical use cases.
- Extracted and processed online data, applied AI/ML models, and improved data-storage workflows to produce usable insights.
- Collected, Preprocessed, Analyzed, and visualized data for real-world applications.

IoT Software & Hardware Engineering - Field Training

June 2025 – Aug 2025

- Completed a seven-week field-training program in embedded systems, cloud connectivity, and IoT architecture, building a foundation for connected-device development.

SKILLS

- Programming Languages:** Python, Java, MySQL, XML, C/C++.
- Data Science:** Pipelines, statistics, Git, and analytical reasoning.
- Big Data & Machine Learning:** AWS (EC2), Scikit-learn, AI tools, large language models, TensorFlow, Rag system, NLP, Database.
- Soft Skills:** Communication, team collaboration, motivation, independent work, and documentation.

PROJECTS

Sales Forecasting for Retail Store

Team Leader

June 2025

- Led a three-person team to build a retail forecasting system with SARIMA, Prophet, and LSTM, delivering six-month forecasts in Streamlit.
- Evaluated model performance, trends, seasonality, and holidays to support forecasting-model selection.
- Tools:** Tracked experiments in MLflow and visualized results with pandas, Seaborn, Plotly, and Matplotlib for stakeholders.

Capstone Project :IoT Smart Pet Care System

Aug 2025

- Built an ESP32 (C++) pet-feeding system with IR and water-level sensors to monitor feeding and fluid needs.
- Integrated hardware with Supabase through MQTT (HiveMQ) for real-time device communication and data exchange.
- Designed a Flutter interface to visualize sensor data and enable remote servo-motor control.
- Tools:** ESP32, C++, MQTT, PlatformIO, Flutter, and Supabase to deliver an integrated pet-care system.

Clinic Rag Bot - Medical AI Assistant

July 2026

- Improved trusted-source retrieval quality to 0.90 similarity by building a PubMedBERT + ChromaDB RAG pipeline restricted to verified medical sources.
- Enabled reliable online and offline assistant operation by integrating Groq and local Qwen2.5 0.5B models into an asynchronous Telegram deployment.
- Reduced unsafe responses by implementing symptom screening and urgent-care escalation guidance.
- Tools :** ChromaDB , PubMedBERT , HuggingFace Transformers , Groq API , Telegram Bot API , telegram-bot , asyncio, SentenceTransformers, NumPy

Job-Market-Insights-analysis

June - 2025

- Developed an interactive Streamlit job market analytics app for Egypt & Saudi Arabia, analyzing 8,000+ real-time job listings to uncover seasonal hiring trends and top 10 high-growth sectors.
- Automated data collection, scraping, and preprocessing, Created 12 interactive visualizations to highlight job distribution, top employers, job types, experience requirements, and market comparisons, Cleaned, transformed, and structured raw data to ensure accurate, reliable insights for users.
- Tools:** Python (Pandas, Regex, BeautifulSoup, SQL Server, Plotly, Matplotlib, Seaborn).

Applied-ML-Projects

- Built 4 applied Machine Learning projects covering Regression, KNN, SVM, and classification, implementing concepts from Hands-On Machine Learning with Scikit-Learn & TensorFlow.
- Optimized KNN classification using GridSearchCV, identifying K=7 and achieving 83% test accuracy across 5,501 test samples.
- Developed an SVM classifier achieving 98.25% test accuracy, with 98% macro F1-score on a 114-sample test set.
- Investigated suspicious model performance using shuffled-label testing and 5-fold Stratified Cross-Validation, obtaining 96.9% mean macro recall ($\pm 2.4\%$) on the original labels versus ~62–63% accuracy with randomized labels.
- **Tools:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebook

DimensionReduction-App

- Reduced 561 $\rightarrow$ 110 features by applying variance/correlation filtering and Group-wise PCA, achieving an 80.4% dimensionality reduction while retaining 96%+ explained variance.
- Improved SVM CV accuracy from 96.22% $\rightarrow$ 97.62% by optimizing feature-reduction parameters and SVM hyperparameters with Optuna, gaining 1.4 percentage points.
- Achieved 92.20% hold-out test accuracy by validating the optimized SVM pipeline on 2,947 unseen samples, demonstrating strong generalization.
- **Tools:** Python, Scikit-learn, Pandas, NumPy, Optuna, PCA, SVM, Matplotlib, Seaborn, Jupyter Notebook, Streamlit app

HONORS AND AWARDS

ECPC (certification)

Competitive Programming

- Placed 93rd out of 146 competing teams representing 5 university after advancing through qualification round.

VOLUNTEER EXPERIENCE

Eahlashbab and Twaheed & Nour

Team leader

Alexandria, Egypt
April 2023 - June 2023

- Led Ramadan food distribution initiatives, coordinating volunteers and ensuring efficient operations, Communications and meal around 15 individuals in day.
- Managed younger volunteers, organized schedules, and oversaw meal delivery logistics.